

Kidney Allocation under Observational Learning

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Motivation

22%

of US
deceased-donor
kidneys are
discarded each
year

90k +

people on the US
national kidney
transplant
waiting list

30%

of waitlist
patients die or are
removed from list
before receiving a
transplant

Why discards happen:

How current US allocation works:

1. A kidney becomes available and is viable for h hours (target: transplant within 18 hours)
2. The national waiting list (operated by OPTN) gives a ranked priority order
3. Offer to patient #1 on waiting list, with a 1 hour response deadline
4. If they decline, offer to patient #2 and so on sequentially
5. After viability window has passed, the kidney is discarded.

Cascading Effect:

- When the patient ahead declines, subsequent patients may also decline
- Each “no” makes the next “no” more likely.
- Observational learning compounds across positions as the kidney moves down the list.

Can a better mechanism fix this?

- 01** Can we reduce kidney discards?
Under realistic decline behavior, can we get closer to omniscient allocation?
- 02** Who would receive the saved kidneys?
If a mechanism saves marginal kidneys, where do they go in the queue? Are front-of-queue patients worse off?
- 03** Is the current system too coarse?
Current OPTN uses a single binary opt-in for marginal kidneys. Is this too coarse?

Model: Patient Response

$$\text{If: } \theta^* > \theta \quad P(\text{accept}) = 0$$

$$\text{If: } \theta^* \leq \theta \quad P(\text{accept}) = \sigma(\alpha + \beta(n-1) - \delta k)$$

θ^* : Patient private reservation threshold

θ : Kidney Quality (1 is best, 0 is worst), based on Kidney Donor Profile Index (KDPI)

n : patient's queue position when offered

α : $\text{logit}(0.9)$, baseline acceptance is 90%

β : 0.10, measure of desperation raising acceptance later in the queue

δ : strength of cascading decline effect

Methods Compared

Omniscient (first-best)

Planner observes every θ_i^* . Assign θ to the highest-priority i with $\theta_i^* \leq \theta$; accept w.p. 1. Discard only if no eligible patient.

Benchmark.

Sequential (one-at-a-time)

Offer down the priority order, one per period, up to H . Patient at position n with k prior declines accepts w.p. $\sigma(\alpha + \beta(n-1) - \delta k)$. Each decline sets $k \leftarrow k+1$, so the **decline signal cascades** down the queue.

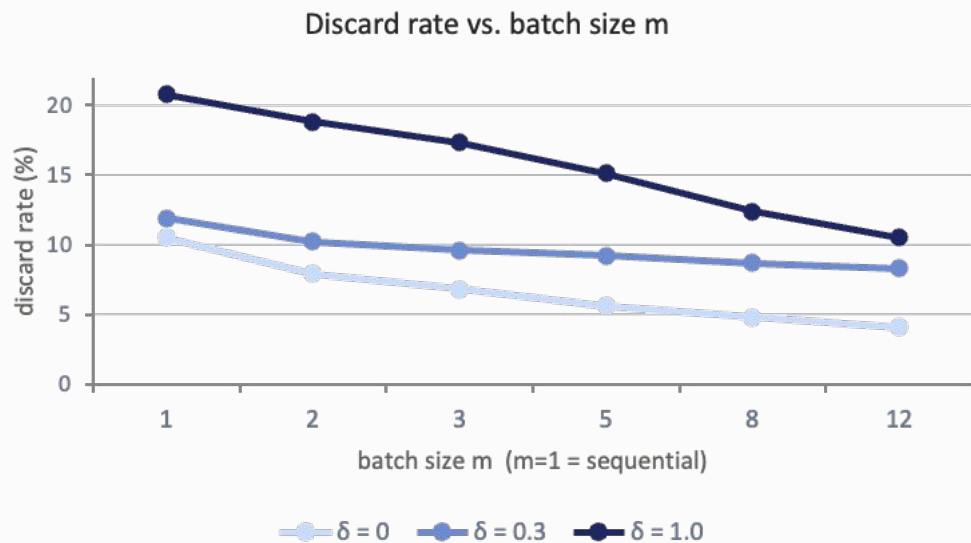
Batching ($m = 5$)

Offer to m patients per round; within a round k is **frozen, so** no one sees a same-round decline. Highest-priority accepter wins; k updates only between rounds. Information-freezing blunts the cascade effect.

Threshold (K buckets)

Patients pre-declare a bucket b_i . θ is offered only to eligible patients ($b_i \leq b(\theta)$), in priority order, up to H . $K = \infty \Rightarrow$ offer only to those with $\theta_i^* \leq \theta$.

Discard Rate: Batching Analysis



m	Discard	Transplant	Mean place
1 (seq)	20.8%	46.7%	1.5
2	18.8%	47.9%	1.5
3	17.3%	48.7%	1.6
5	15.1%	50.1%	1.7
8	12.4%	51.7%	1.9
12	10.5%	52.8%	2.2

- Wider simultaneous offers always help: discard rates fall monotonically at every δ .
- Batching pays off most when δ is high, since larger batch sizes freeze the decline signal.
- Doesn't fully escape: k still updates across batches.

Results: Strong Observational Learning

Mechanism	Discard	Mean placement	Transplant rate	Mean wait
Omniscient (first-best)	2.3%	22.2	47.4%	505
Sequential	19.6%	1.5	32.7%	658
Batching (m = 5)	14.4%	1.7	37.2%	613
Threshold K=2 (OPTN)	15.1%	3.8	36.6%	621
Threshold K=5	10.7%	2.7	40.4%	581
Threshold continuous (K= ∞)	2.6%	19.6	47.2%	508

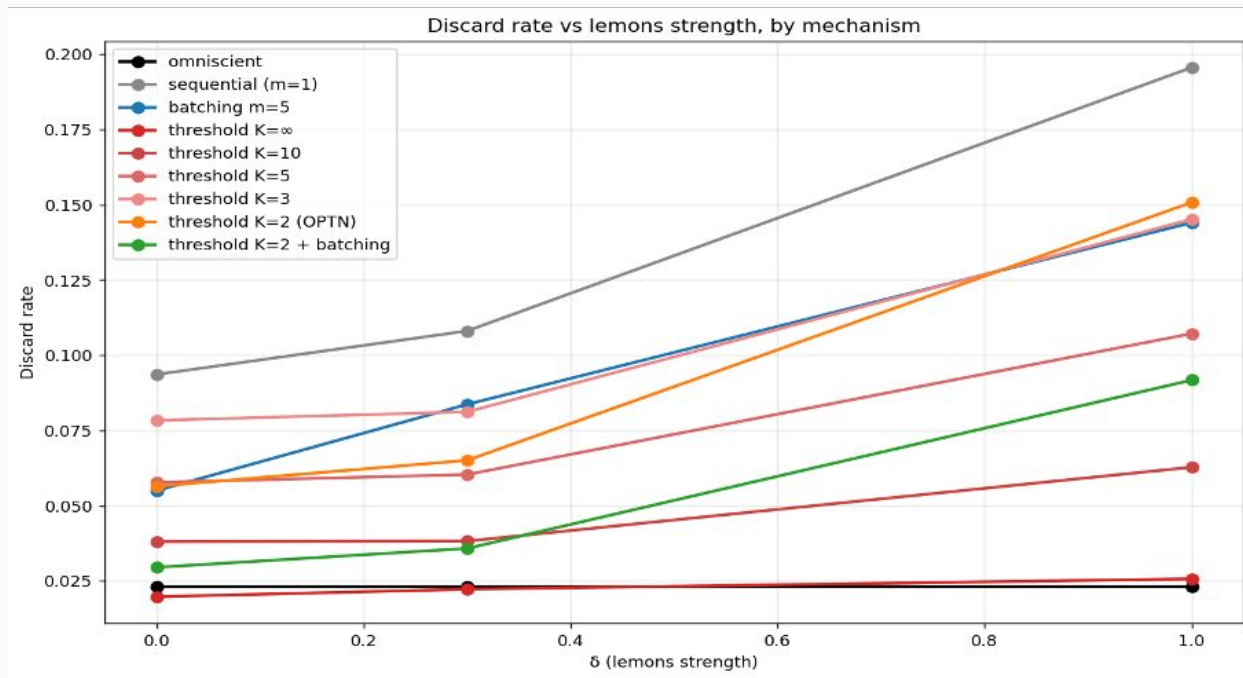
- Mean placement = average queue rank of the recipient (lower = nearer the front)
- Mean wait = kidneys arriving between a recipient joining and being matched
- $\delta = 1.0$.

Results: Varying Observational Learning

Mechanism	$\delta = 0$	$\delta = 0.3$	$\delta = 1.0$	Δ over sweep
Omniscient (first-best)	2.3%	2.3%	2.3%	0.0 pp
Sequential	9.4%	10.8%	19.6%	+10.2 pp
Batching (m = 5)	5.5%	8.4%	14.4%	+8.9 pp
Threshold K=2 (OPTN)	5.6%	6.5%	15.1%	+9.5 pp
Threshold K=5	5.8%	6.0%	10.7%	+4.9 pp
Threshold K=10	3.8%	3.8%	6.3%	+2.5 pp
Threshold continuous (K= ∞)	2.0%	2.2%	2.6%	+0.6 pp

- Self-selection insulates changes in the cascade effect.
- Less-coarse threshold methods move less than sequential or batching.

Discard Rates: as cascading effect worsens

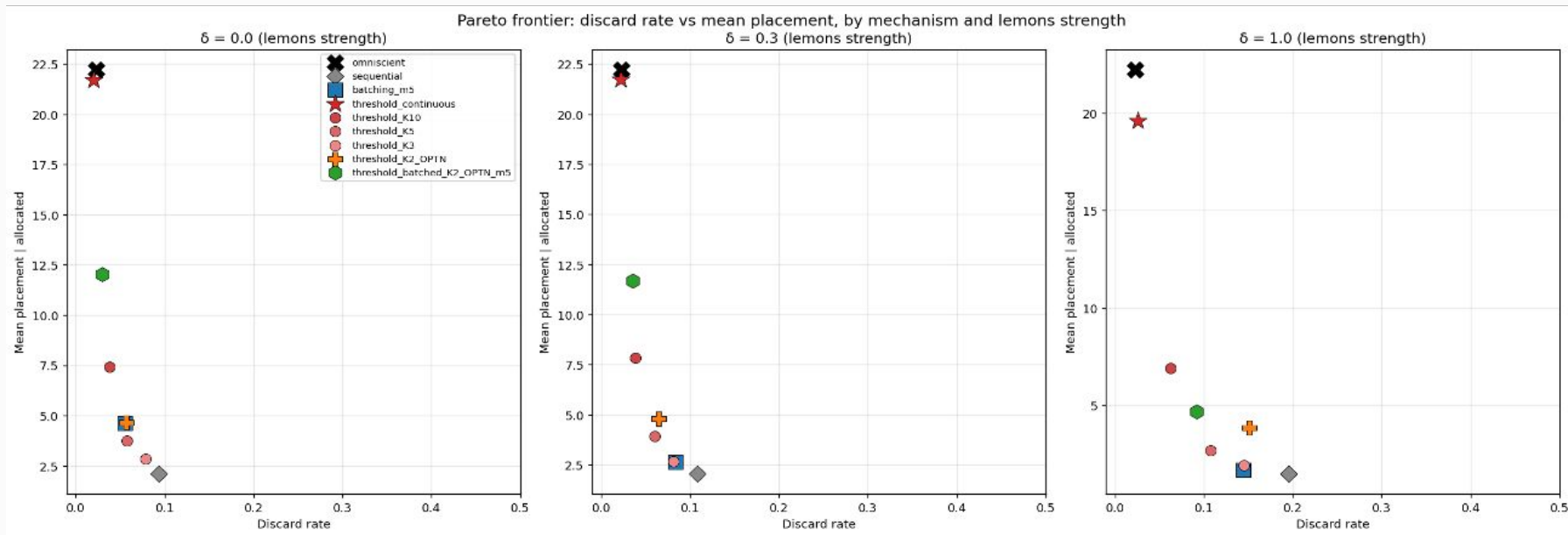


X-axis: strength of cascading decline effect (δ)

Y-axis: discard rate

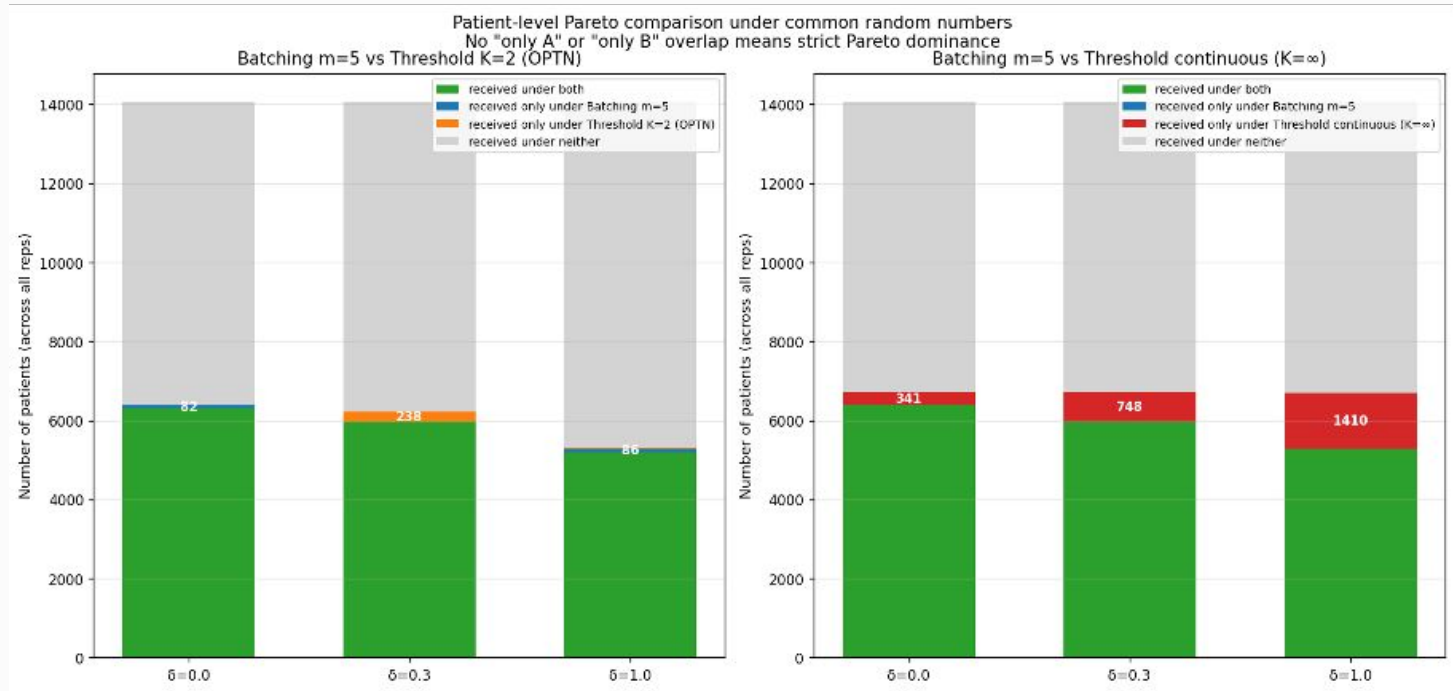
- Sequential and batching discard rates rise sharply as δ increases.
- Continuous threshold stays at the omniscient performance level.
- Coarser thresholds show improvement over sequential.

Discards vs. Allocation Depth



- As δ grows, sequential and batching show more discards while threshold stays closer to omniscient

Is anyone worse off?



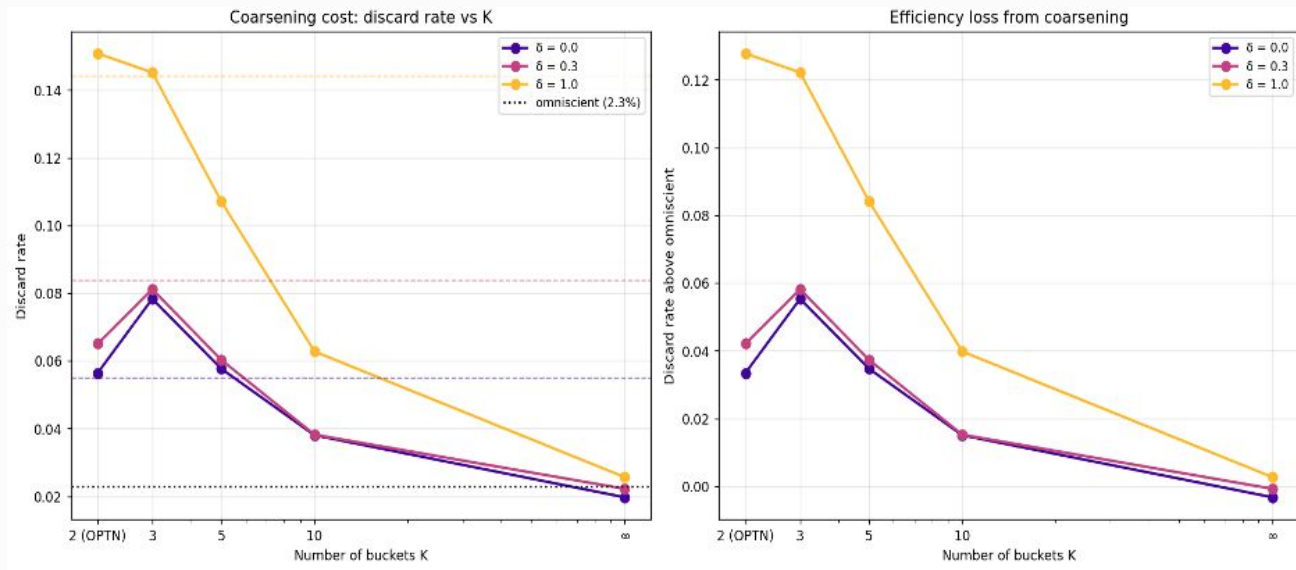
- From simulations: threshold (continuous scale) is strictly better than batching. More patients better off, zero worse off.
- Saved kidneys are those that were discarded in batching, not redirected from front-of-queue patients.

What about more coarse thresholds?

Pairing (better under threshold / under greedy)	$\delta = 0$	$\delta = 0.3$	$\delta = 1.0$
K = 2 (OPTN) vs Sequential	479 / 0	564 / 0	650 / 0
K = 2 (OPTN) vs Batching	1 / 113	255 / 0	20 / 65
K = 10 vs Sequential	753 / 0	910 / 0	1,724 / 0
K = 10 vs Batching	162 / 0	601 / 0	1,029 / 0

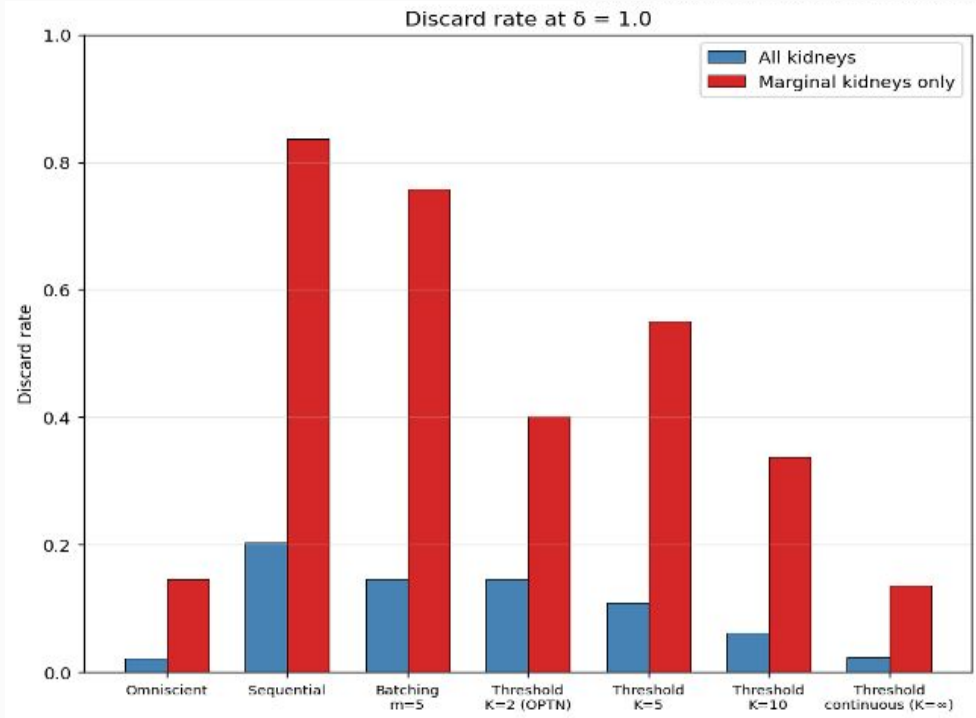
- K=10 dominates everything: the more detail we can get from self-reported preferences, the better off kidney recipients are. Patients are never worse off in this regime compared to sequential or batching.
- K=2 beats sequential but not batching.
- General takeaway: more granularity improves performance.

Threshold Coarseness



- Cost of coarseness grows as the strength of cascading decline effect δ grows.
- Going from $K=2$ to $K=\infty$ saves 12.5% at $\delta=1.0$ but 3.6% at $\delta=0$.
- $K=3 < K=2$: Uniform cuts miss the threshold distribution mass (doesn't really cut off where it matters)
- OPTN $K=2$: Cuts off where it matters (KDPI 85)

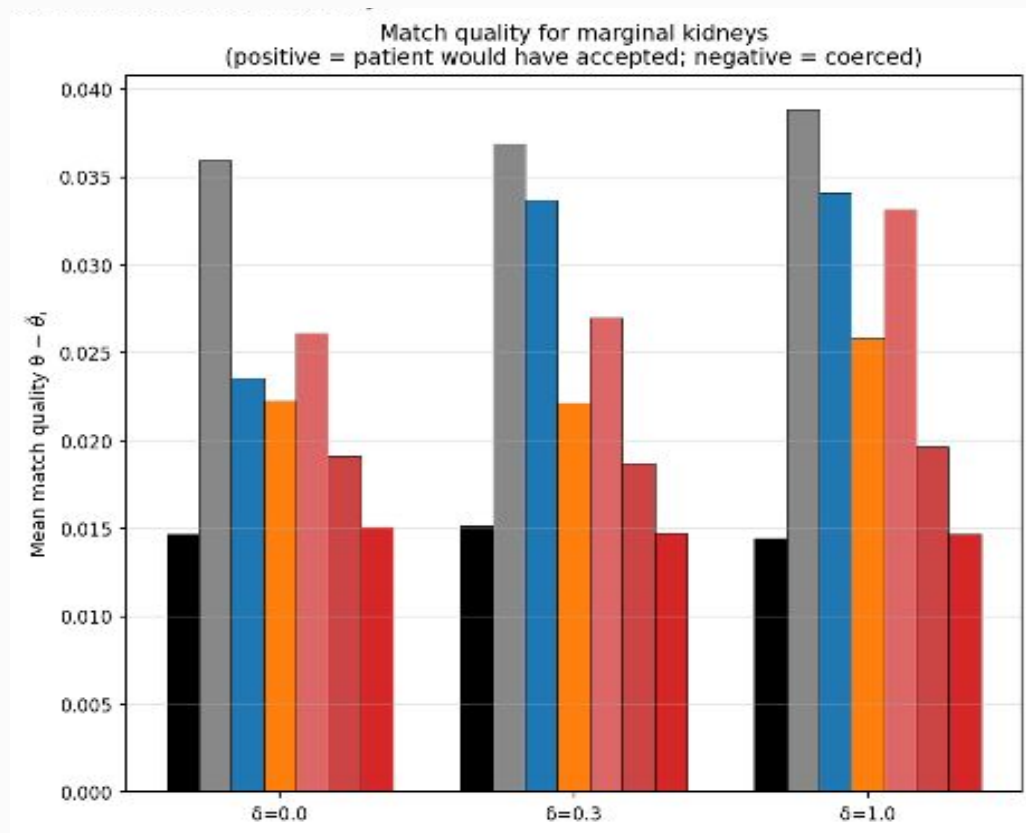
Marginal Kidneys: Lowering Discard Rates



When the cascading decline effect is high:

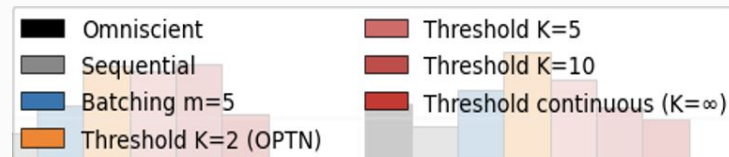
- Sequential discards >80% of marginal kidneys because they never reach patients who would accept.
- Batching does slightly better by improving queue penetration
- Continuous threshold discards about 14%, which is similar to omniscient.
 - Other threshold methods do better than sequential and batching.
 - Generally, improving granularity does improve the discard rate EXCEPT for K=2 at the current OPTN question level.
 - Threshold K=2 (OPTN) is based on the current OPTN question value of KDPI 85.

Marginal Kidneys: Where do they go?

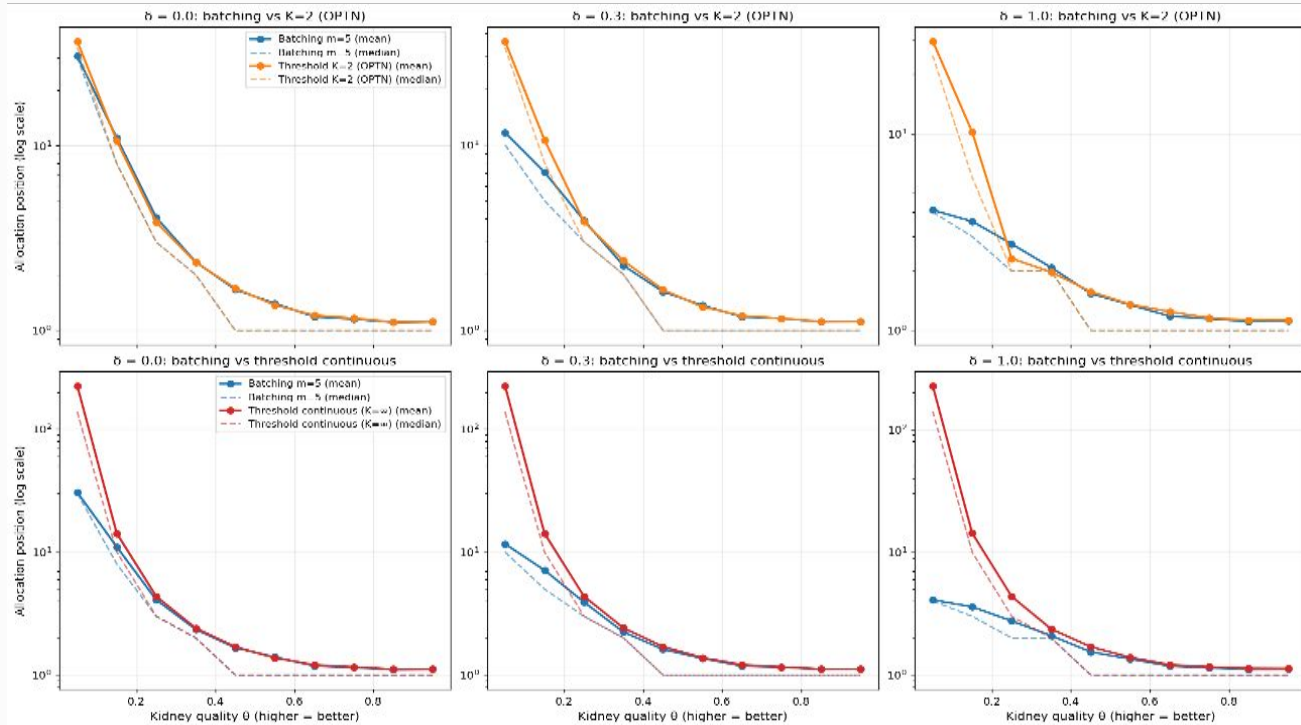


Did the marginal kidneys go to the people that wanted them?

- Yes, match quality is positive everywhere.
- No one is coerced.
- Threshold mechanism and omniscient match more tightly because they have smaller surplus per allocation. However, they have more allocations/lower discard rate.

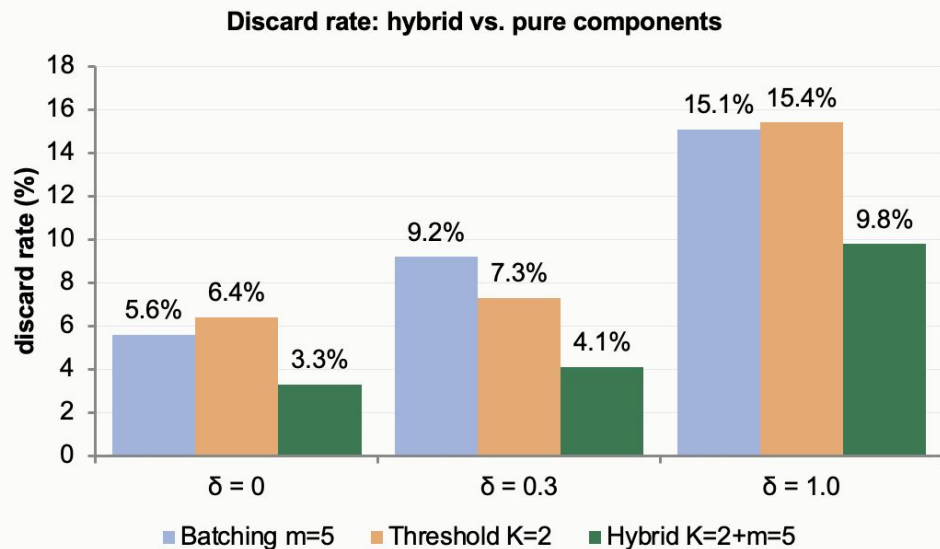


Marginal Kidneys: Queue Position



- Mean queue position at allocation as a function of kidney quality.
- Premium kidneys (right side of the axis) have very low position allocation (as expected).
- For marginal kidneys, threshold continuous walks to position 100+ to find an acceptor.
- Guaranteed that the people previously in line wouldn't have accepted or had a chance to decline.

Combining Batching + Threshold



Mechanism ($\delta = 1.0$)	Discard	Transplant	Mean place
Batching m=5	15.1%	50.1%	1.7
Threshold K=2	15.4%	49.9%	4.1
Hybrid K=2+m=5	9.8%	53.2%	5.4
Threshold K=10	6.6%	55.2%	7.9
Hybrid K=10+m=5	3.4%	57.0%	18.6

- Hybrid helps strictly more patients than batching (637/0 at $\delta=1.0$) and than threshold alone (676/0 at $\delta=1.0$)
- Raises transplant rate and lowers discard rate. The mean placement gets worse due to reaching deeper in the queue.
- Intuition: threshold filter stops offers to ineligible patients, and batching continues to freeze the decline signal within the batch (assuming the number of previous batches is known).

Summary

1. Threshold continuous matching is basically omniscient and eliminates the rejection cascade. However, it's not a realistic solution:
 - a. Difficult for patients to give a cutoff of which KDPI scores they would or would not accept.
 - b. Buckets (more coarseness) is better for understanding.
2. $K=10$ (granular) threshold method dominates batching and sequential. No patient is worse off, many are better off.
3. Threshold mechanisms have to reach deep into the queue for marginal kidneys. However, highly desirable kidneys still go to the front of the queue.
4. Coarse threshold ($K=2$ OPTN) captures most of the gain already, and finer buckets of $K=5$ and $K=10$ capture more gain, especially at high δ .
5. Batching + Threshold complement each other and hybrid outperforms single methods.

Limitations

1. Zhang (2010) argues that the rejection cascade is real: observational learning drives declines cascade. This is where the threshold method findings would have the most impact.
2. Agarwal et al (2021) argue that declines reflect private kidney characteristics, not learning. Then, these private kidney characteristics are not modeled in our described case.
3. Next steps: real-world panel data on offer responses.

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Appendix

Model Calibrations (Based on real data)

Probability patient i accepts kidney θ

$P(\text{accept} \mid n, k) =$

0 if $\theta^*_i > \theta$

$\sigma(\alpha + \beta(n-1) - \delta k)$ if $\theta^*_i \leq \theta$

Object	Calibration	Anchor
Kidney quality θ	Uniform[0,1]	1 – KDPI/100
Patient threshold θ^*	Beta(2,5)	~30% marginal
Baseline accept $p(\theta)$	$\sigma(-3.4 + 5.7\theta)$	Mohan 2018
Lemons strength δ	{0, 0.3, 1.0}	Zhang 2010
Load p	1.6 / kidney	US DD ratio
Offer budget H	12 / kidney	~12h window
Replications	15 runs, CRN	variance ctl

More Formal Method Definition

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Benchmark.

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Offer down the priority order, one per period, up to H . Patient at position n with k prior declines accepts w.p.

$\sigma(\alpha + \beta(n-1) - \delta k)$. Each decline sets $k \leftarrow k+1$, so the **decline signal cascades** down the queue.

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Threshold (K buckets)

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Where the acceptance equation comes from

$$P(\text{accept}) = 0 \text{ if } \theta^* > \theta; \text{ else } \sigma(\alpha + \beta(n-1) - \delta k)$$

The threshold gate ($\theta^* > \theta \rightarrow \text{reject}$) — reservation-quality / optimal-stopping. Patients hold a private cutoff and reject anything below it.

Source: McCall (1970) search theory; Su & Zenios (2004, 2005) organ-acceptance thresholds.

The logistic acceptance $\sigma(\alpha + \beta(n-1) - \delta k)$ — random-utility / binary logit. α sets the 90% baseline; β raises acceptance deeper in the queue (desperation).

Source: McFadden (1974) discrete choice; Agarwal, Ashlagi, Rees, Somaini & Waldinger (2021) sequential organ assignment.

The cascade term $-\delta k$ — observational learning. Each prior decline lowers the next patient's odds; linear in $k = \text{i.i.d.-signal Bayesian updating}$.

Source: Banerjee (1992) and Bikhchandani, Hirshleifer & Welch (1992) informational cascades; Zhang (2010) kidney declines.

- Each component is from literature: linear-in- k in cascade is a reduced form, so we use the sweep over $\bar{\delta}$ as a robustness check